Shane Woloszyn

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Education

Purdue University — West Lafayette, IN
Bachelor of Science in Computer Science and Economics — *Aug. 2025 – May 2028*

- **Activities:** John Martinson Honors College • BuildPurdue • Purdue Magnetics Lab • Sigma Alpha Epsilon Fraternity – Recruitment Chair • Certificate in Entrepreneurship and Innovation
- **Relevant Coursework:** Discrete Math, Programming in C, Linear Algebra, Object-Oriented Programming, Computer Architecture, Data Structures and Algorithms, Multivariate Calculus

Experience

Software Engineering Intern — May 2026 – Present
Liberty Mutual Insurance Group — *Portsmouth, NH*

- Contributing to Twilio Flex enterprise contact center platform supporting 20k+ concurrent agents, with work spanning real-time call handling, compliance workflows, and performance analytics
- Developing backend logic and Typescript/React UI components for a Fair Credit Reporting Act disclosure system handling 4% of eligible calls at a 10M+ monthly call contact center

Co-Founder and Lead Developer — Dec. 2025 – Present
Liora — *West Lafayette, IN*

- Founded and led engineering for a hotel management platform serving independent properties, owning architecture, technical hiring, and product direction from day one
- Built the initial product in Java with Spark and Twilio, shipping working prototypes for early customer demos within the first month
- Selected for VentureX accelerator at Purdue, receiving $5k funding

Independent Researcher — June 2024 – Feb. 2025
University of New England — *Biddeford, ME*

- Designed and implemented a Conditional Generative Adversarial Network (CGAN) using Python, PyTorch, and NumPy for generating synthetic chest X-ray images aimed at improving medical education.
- Validated outputs statistically against ground-truth CT data and presented results at the Summer Undergraduate Research Symposium under Dr. Sylvain Jaume.

Projects

Crypto Arbitrage Trading Dashboard | *Python, Streamlit, CCXT*

- Built a real-time arbitrage detector that pulls live order book data from 5+ exchanges via CCXT and flags fee-aware profit opportunities
- Backtested strategies on simulated paper trades, tracking cumulative returns and potential returns across thousands of trades

File Loading Operating System | *C, x86 Assembly, Git*

- Developed a simple operating system bootloader and kernel supporting basic file read, write, and edit functionality in a simulated RAM disk environment
- Implemented low-level VGA text mode output and PS/2 keyboard input handling using direct hardware port communication
- Programmed the transition from real mode to protected mode on x86 architecture, including Global Descriptor Table (GDT) setup and segment register configuration

Technical Skills

Languages: Python, Java, C, TypeScript, JavaScript, x86 Assembly
Frameworks & Tools: React, PyTorch, NumPy, Unity, Spark, Git, Unix/Linux, Twilio, REST APIs
Concepts: Machine learning, distributed systems, object-oriented design, Agile/Scrum, data analytics, algorithmic trading